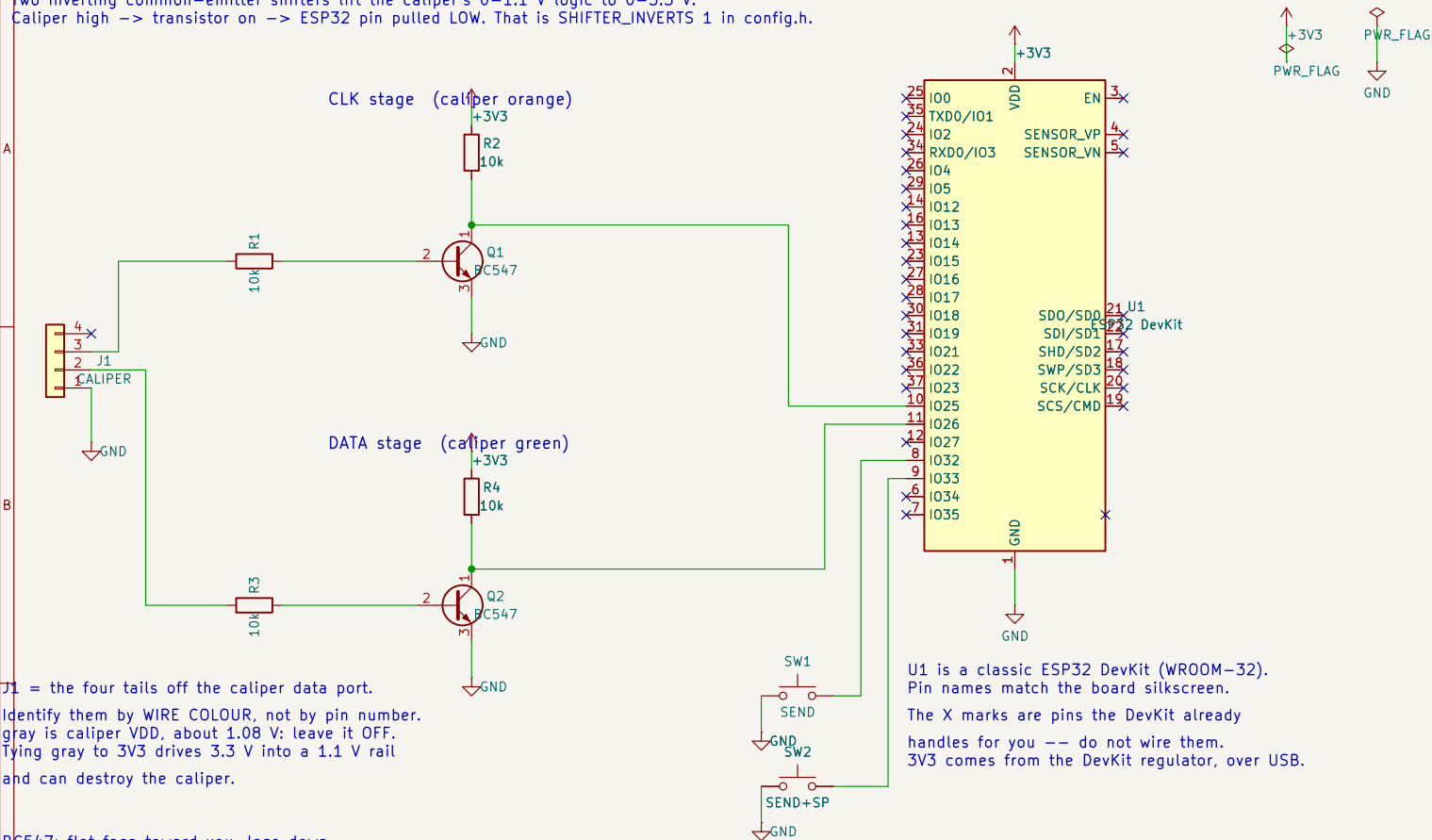


Two inverting common-emitter shifters lift the caliper's 0–1.1 V logic to 0–3.3 V. Caliper high $\rightarrow$ transistor on $\rightarrow$ ESP32 pin pulled LOW. That is SHIFTER_INVERTS 1 in config.h.



11 = the four tails off the caliper data port.
Identify them by WIRE COLOUR, not by pin number.
gray is caliper VDD, about 1.08 V: leave it OFF.
Tying gray to 3V3 drives 3.3 V into a 1.1 V rail
and can destroy the caliper.

BC547: flat face toward you, legs down --
left = C = pin 1, middle = B = pin 2,
right = E = pin 3.

A 2N3904 is the mirror image (E B C), so check
before soldering: diode mode, red on the middle
leg, both outer legs read -0.7 V . The HIGHER of
the two is the EMITTER.

SW1/SW2 are for BLE mode only -- not needed for CALIPER_SNIFFER_MODE bring-up.

U1 is a classic ESP32 DevKit (WROOM-32).
Pin names match the board silkscreen.
The X marks are pins the DevKit already
handles for you -- do not wire them.
3V3 comes from the DevKit regulator, over USB.

Sheet: /		D
File: protoboard.kicad_sch		
Title: BLE Digital Caliper -- protoboard prototype		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 1/1